

Routing Algorithms

Subject: Computer Networks Topic: Network Layer

Overview

Distance vector, link state, path vector routing with examples

Routing problem

- A router selects a next hop for a destination using a routing table and a route metric.
- Link-state protocols distribute topology information and compute shortest paths locally.
- Distance-vector protocols exchange reachability and costs with neighbors.

Algorithms

- Dijkstra's algorithm repeatedly finalizes the closest unsettled vertex and relaxes its outgoing edges.
- Bellman-Ford uses repeated edge relaxation and can handle negative edge costs; networking protocols add safeguards against instability.
- BGP is a path-vector protocol that exchanges reachability between autonomous systems and applies policy, not merely shortest distance.

Worked shortest-path example

- For edges $A-B=2$, $A-C=5$, $B-C=1$, Dijkstra from A first sets $B=2$ and $C=5$.
- Processing B improves C to 3 via $A-B-C$; record B as C's predecessor.
- The path is reconstructed by following predecessor links backward from the destination.

Convergence and loops

- Distance-vector protocols can suffer count-to-infinity after failures; split horizon and poisoned reverse reduce some loops.
- Link-state flooding and recomputation help converge after topology changes but require memory and CPU.
- Administrative policy, equal-cost paths, and failure detection affect real route selection.